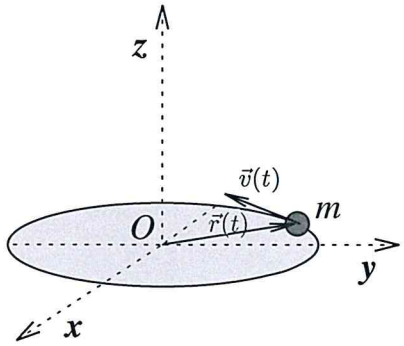
Problem (1) (25 points)

A particle of mass m moves in its elliptical orbit as shown in the figure. The position vector of the particle is given as follows:

$$\vec{r}(t) = R\sin(\omega t)\ \hat{\imath} + 2R\cos(\omega t)\ \hat{\jmath}$$



(a) (6 pts.) Find the velocity of the particle as a function of time.

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = R\omega\cos(\omega t)\hat{\imath} - 2R\omega\sin(\omega t)\hat{\jmath}$$

(b) (6 pts.) Find the acceleration of the particle as a function of time.

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} = \frac{d\vec{v}}{dt} = -R\omega^2\sin(\omega t)\hat{\imath} - 2R\omega^2\cos(\omega t)\hat{\jmath}$$

(c) (6 pts.) Find the angle between the velocity and the acceleration at $t = \frac{\pi}{2\omega}$.

$$\vec{v}(t)\cdot\vec{a}(t) = |v|\,|a|\cos\theta\ ;\quad \vec{v}\left(\tfrac{\pi}{2\omega}\right) = -2R\omega\hat{\jmath} \quad \vec{a}\left(\tfrac{\pi}{2\omega}\right) = -R\omega^2\hat{\imath}$$

$$(-2R\omega\hat{\jmath})\cdot(-R\omega^2\hat{\imath}) = (2R\omega)(R\omega^2)\cos\theta$$

$$0 = 2R^2\omega^3\cos\theta \rightarrow \cos\theta = 0\ ;\quad \theta = \frac{\pi}{2}$$

(d) (7 pts.) Find the angular momentum of the particle with respect to point O. Is there an external torque on this particle?

$$\vec{L} = \vec{r}\times\vec{p} = \vec{r}\times(m\vec{v}) = m(\vec{r}\times\vec{v})$$

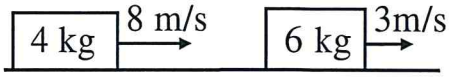
$$\vec{L} = m\left[(R\sin(\omega t)\hat{\imath} + 2R\cos(\omega t)\hat{\jmath})\otimes(R\omega\cos(\omega t)\hat{\imath} - 2R\omega\sin(\omega t)\hat{\jmath})\right]$$

$$\vec{L} = m\left[-2R^2\omega\sin^2(\omega t)\hat{k} - 2R^2\omega\cos^2(\omega t)\hat{k}\right]$$

$$\vec{L} = m\left[-2R^2\omega\right]\hat{k}\ ;\quad \frac{d\vec{L}}{dt} = 0$$ so there is no external torque

Problem (2) (25 points)

A mass m_1=6 kg is moving to the right with a velocity of 3 m/sec. Another mass m_2= 4 kg is moving to the right with a velocity of 8 m/sec. There is <u>no friction</u> between the objects and the ground. Find the final velocities of m_1 and m_2:



(a) (10 pts.) If they make a <u>completely inelastic</u> collision.

Momentum is conserved. $P_{initial} = P_{final}$

$P_{1\,initial} = 4 \cdot 8 = 32 \text{ kg}\frac{m}{s}$

$P_{2\,initial} = 6 \cdot 3 = 18 \text{ kg}\frac{m}{s}$

$P_{initial} = (32+18) = 50 \text{ kg}\frac{m}{s}$

$P_{final} = (4+6)\, V_f$ Because they move together.

$V_f = 5 \text{ m/s}$

(b) (15 pts.) If they make an <u>elastic</u> collision.

$P_{initial} = 50 \text{ kg}\frac{m}{s}$

From momentum conservation; $P_{initial} = P_{final}$

Combine these 2 eqns.

$50 = 4 \cdot V_{1f} + 6 \cdot V_{2f}$

Also using $V_{1i} + V_{1f} = V_{2i} + V_{2f}$

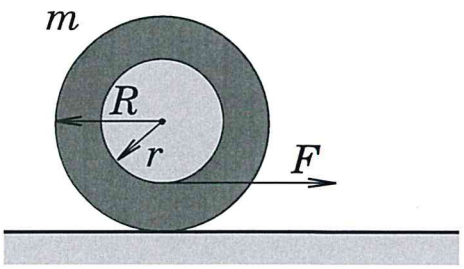
$8 + V_{1f} = 3 + V_{2f} \rightarrow V_{1f} - V_{2f} = -5$

$4 V_{1f} + 6 V_{2f} = 50$

$+\ 6(V_{1f} - V_{2f}) = -30$ (multiply with 6)

$10 V_{1f} = 20$ $V_{1f} = 2 \text{ m/s}$ $V_{2f} = 7 \text{ m/s}$

Problem (3) (25 points)



A disk yo-yo with mass m and radius $R = 2r$ it being pulled by a force with magnitude F from a distance r away from its center of mass as shown in the figure. Then, the yo-yo starts to roll without slipping.

(Your answers must be in terms of m, r, F and g.

Rotational inertia of a disk about an axis through its center $I = \frac{1}{2}mR^2$.)

(a) (12 pts.) Find the acceleration of the yo-yo.

(*Hint: Think about the torque with respect to the point where the disk touches to the ground. This will tell you the direction of its motion and rotation.*)

$\longrightarrow F$

$\longleftarrow F_f$

Direction of motion is $\rightarrow$

Also using Newton's 2nd Law

$F - F_f = ma = m.(\alpha . \overset{2r}{R})$ (2)

$\tau = I\alpha$

$F_f . R - F.r = \frac{1}{2} m R^2 \alpha$

$2F_f . r - F.r = \frac{1}{2} m 4r^2 \alpha$

$2F_f . r - F.r = 2mr^2 \alpha$

$2F_f - F = m(\underbrace{2r\alpha}_{a})$

$2F_f - F = ma$ (1)

Combine (1) and (2)

$F_f = 2ma \qquad F = 3ma$

$$\boxed{a = \frac{F}{3m}}$$

$a=$

(b) (13 pts.) Find the friction applied to the yo-yo.

Combine (1) and (2)

$2F_f - F = ma$

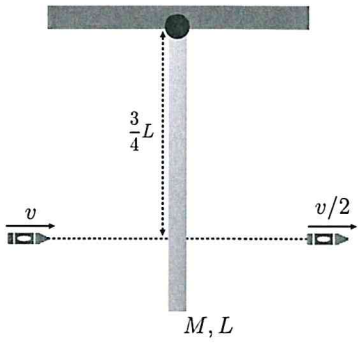
$F - F_f = ma$

$F_f = 2ma \qquad F = 3ma$

$$\boxed{F_f = \frac{2}{3} F}$$

$f=$

Problem (4) (25 points)



A bullet with mass m and speed v penetrates through a rod with mass M and length L that is fixed at its one end with a frictionless hinge as shown in the figure. The bullet's speed is observed to be $v/2$ after the penetration.

(Your answers must be in terms of m, M, L, v and g.

Rotational inertia of a rod about and axis through one of its end $I = \frac{1}{3}ML^2$.))

(a) (10 pts.) Find the angular speed of the rod just after the bullet leaves it.

From conservation of angular momentum;

$L_{initial} = L_{final}$

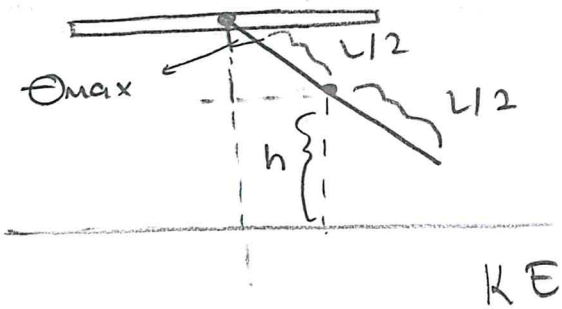
$$m.V.\frac{3}{4}L = m.\frac{V}{2}.\frac{3}{4}L + I\omega$$

$$\frac{3}{4}mVL - \frac{3}{8}mVL = \frac{1}{3}ML^2.\omega$$

$$\frac{3}{8}mVL = \frac{1}{3}ML^2\omega \rightarrow \omega = \frac{9}{8}\frac{V}{L}\frac{m}{M}$$

$\omega=$

(b) (15 pts.) Find the maximum angle θ_{max} that the rod will make with the vertical.



$$h = \frac{L}{2} - \frac{L}{2}\cos\theta$$

Just after the collision

Rotational Kinetic energy of the rod;

$$KE_{rot} = \frac{1}{2}I\omega^2 = \frac{1}{2}.\left(\frac{1}{3}ML^2\right).\left(\frac{9}{8}\frac{V}{L}\frac{m}{M}\right)^2$$

$$KE_{rot} = \frac{27}{128}\frac{m^2V^2}{M}$$

$\theta_{max}=$

when the rod make θ_{max} angle with the vertical its potential energy $U = M.g.\left(\frac{L}{2} - \frac{L}{2}\cos\theta\right)$

$$KE_{rot} = U \rightarrow \frac{27}{128}\frac{m^2V^2}{M} = M.g.\frac{L}{2}(1-\cos\theta)$$

$$\frac{27m^2V^2}{64gLM^2} = 1-\cos\theta \rightarrow \cos\theta = \left(1 - \frac{27V^2m^2}{64gLM^2}\right)$$

$$\theta_{max} = \text{Arc}\cos\left(1 - \frac{27V^2m^2}{64gLM^2}\right)$$